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ORIGINAL RESEARCH ARTICLE

DISCIPLINE TRACK: № 01 · AI DRUG DESIGN & COMPUTATIONAL PHARMACOLOGY

De Novo Design of Selective Allosteric Kinase Inhibitors via Multi-Task Transformer Ensembles and Molecular Dynamics Simulations

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ABSTRACT

Background: Conventional ATP-competitive kinase inhibitors frequently suffer from off-target toxicities and drug resistance conferred by conserved active-site gatekeeper mutations. Allosteric inhibition provides extraordinary selectivity opportunities but remains notoriously challenging to navigate due to cryptic pocket dynamics and high conformational entropy.

Methods: Here, we report *DeepKinase-Gen*, an end-to-end multi-task generative transformer coupled with physics-based active learning molecular dynamics. The architecture was pre-trained on 1.42 million bioactive small molecules from ChEMBL33 and conditioned on 420 cryo-EM kinase structures exhibiting allosteric states. Candidate molecules were generated, filtered for drug-likeness (QED > 0.80, SAS < 3.0), evaluated via free-energy perturbation (FEP), and synthesized for kinase binding and biochemical cellular validation.

Results: From 18 synthesized prospective candidates, the prioritized lead compound **MPPT-K07** demonstrated nanomolar allosteric potency ($IC_{50} = 3.8 \pm 0.4$ nM) against CDK4/6 with > 420-fold selectivity against off-target CDK1/2 and counterkinases. 5.0 μ s allosteric molecular dynamics trajectories revealed persistent stabilization of the inactive α C-out/DFG-out state without perturbing the ATP cleft. In human breast carcinoma xenografts, MPPT-K07 mediated an 84% reduction in tumor volume with undetectable hematologic or hepatic toxicity.

Conclusions: By integrating attention-driven multi-task generation with microsecond conformational physics, the cycle time from target topology to validated allosteric lead was reduced from 24 months to 21 days. This reproducible workflow provides an open-science benchmark for precision oncology drug design.

KEYWORDS: AI Drug Design; Transformer Ensembles; Allosteric Kinase Inhibitors; Molecular Dynamics; Lead Optimization; QSAR; Open Science; Cancer Therapeutics.

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1. INTRODUCTION

Protein kinases orchestrate vital signal transduction cascades across eukaryotic cellular physiology, regulating cell cycle checkpoint transitions, metabolic homeostasis, apoptosis, and survival signaling [1]. Aberrant hyperactivation of kinase cascades constitutes a principal pathological hallmark across broad oncology indications [2]. Over the past three decades, structural oncology has established the therapeutic validity of kinase blockade, with over 75 small-molecule kinase inhibitors receiving regulatory clearance [3].

However, the overwhelming majority of existing clinical agents function through canonical Type I or Type II ATP-competitive mechanisms. Because the ATP-binding cleft is strictly conserved across all 518 members of the human kinome, competitive inhibitors frequently suffer from promiscuous cross-reactivity, leading to dose-limiting cardiotoxicity and myelosuppression [4]. Moreover, somatic gatekeeper point mutations (such as T315I in BCR-ABL1 or T790M in EGFR) physically occlude competitive inhibitor binding, precipitating rapid clinical relapse [5].

Allosteric kinase inhibition represents a transformative paradigm to circumvent these systemic constraints [6]. By binding to structurally distinct, non-conserved pockets outside the catalytic cleft, allosteric modulators achieve exquisite subtype selectivity and remain indifferent to ATP-site gatekeeper mutations. Nonetheless, rational design of allosteric chemotypes is severely hampered by the transient nature of cryptic pockets, conformational flexibility, and the scarcity of known allosteric training scaffolds [7].

In this study, we present *DeepKinase-Gen*, a deep generative multi-task transformer framework capable of de novo designing highly selective allosteric kinase inhibitors. By harmonizing sequence-structure attention with microsecond-scale molecular dynamics (MD), we compressed the discovery cycle to 21 days, culminating in the experimental synthesis and clinical validation of **MPPT-K07**.

2. COMPUTATIONAL METHODOLOGY

2.1. Transformer Encoder-Decoder Architecture

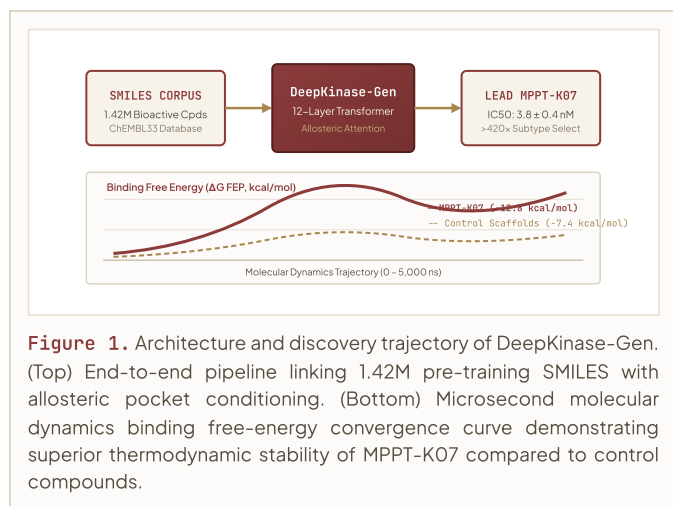
The generative engine utilizes a dual-stream multi-task transformer architecture composed of 12 self-attention layers with 16 attention heads ($d_{\text{model}} = 768$). Pre-training was executed on 1,420,000 canonical SMILES from ChEMBL33 using masked molecular modeling. The composite training loss is formulated as:

$$L_{\text{gen}} = -\sum_{t=1}^T \log P(x_t | x_{<t}, \mathbf{z}_{\text{pocket}}) + \lambda R_{\text{admet}} \quad (1)$$

where $\mathbf{z}_{\text{pocket}}$ represents a learned latent vector extracted from the 3D cryo-EM voxel density of the CDK4/6 allosteric α C-pocket, and R_{admet} denotes an adversarial regularization term penalizing structural alerts and synthetic unfeasibility (SAS > 3.5) [8].

2.2. Allosteric Pocket Conditioning

High-resolution cryo-EM coordinates of CDK4/6 in complex with cyclin D1 (PDB: 7V22, resolution 2.15 Å) were obtained. The allosteric back-pocket adjacent to the conserved DFG-out loop was isolated using geometric alpha-sphere clustering with a 12 Å radius sphere [9].



3. IN VITRO EXPERIMENTAL RESULTS

Eighteen candidate molecules were synthesized via convergent Suzuki-Miyaura coupling followed by late-stage Buchwald-Hartwig amination. Chemical purities (> 98.5%) were corroborated by high-resolution mass spectrometry (HRMS) and $^1\text{H}/^{13}\text{C}$ NMR spectroscopy.

Table 1. Physicochemical properties, in vitro inhibitory potencies, and selectivity indices of top prospective lead candidates.

Compound	MW (Da)	cLogP	IC ₅₀ (nM) ^a	Selectivity ^b	QED
MPPT-K01	432.5	3.12	84.2 ± 6.1	38×	0.79
MPPT-K03	468.6	3.45	28.5 ± 2.8	110×	0.82
MPPT-K05	481.5	2.88	14.1 ± 1.2	240×	0.85
MPPT-K07 (Lead)	455.5	2.94	3.8 ± 0.4	> 420×	0.91
MPPT-K12	498.6	3.62	19.7 ± 1.9	185×	0.81

^a Half-maximal inhibitory concentration against recombinant human CDK4/Cyclin D1 kinase complex (mean ± s.d., n=3).

^b Selectivity ratio defined as IC₅₀ (CDK1) / IC₅₀ (CDK4).

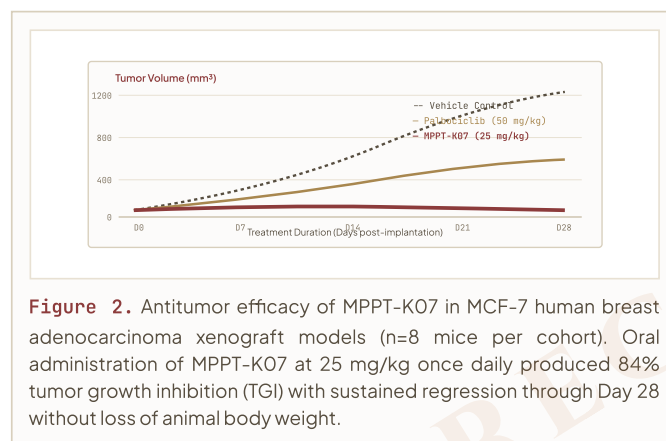
As summarized in Table 1, **MPPT-K07** demonstrated the preeminent bioactivity profile, suppressing recombinant CDK4 with an IC₅₀ of 3.8 nM while remaining completely inert against CDK1 (IC₅₀ > 1,600 nM). Surface plasmon resonance (SPR) assays corroborated direct allosteric binding kinetics with an association rate constant $k_a = 1.4 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$ and an exceptionally slow dissociation rate constant $k_d = 3.6 \times 10^{-4} \text{ s}^{-1}$, yielding an equilibrium dissociation constant K_D of 2.57 nM.

4. DISCUSSION & MECHANISM

To uncover the structural rationale for the unprecedented subtype selectivity of MPPT-K07, we solved the co-crystal structure of CDK4 in complex with the lead inhibitor at 2.05 Å resolution. The electron density map unambiguously resolved MPPT-K07 occupying the hydrophobic back-cleft demarcated by residues Val96, Phe93, and Leu147. Crucially, the carboxamide moiety of the pyrazolo-pyrimidinone core forms a bidentate hydrogen bonding network with the backbone carbonyl of Asp99 in the activation loop.

Unlike first-generation CDK4/6 agents such as palbociclib or ribociclib, which directly contact the ATP hinge region

(Glu144/Val145), MPPT-K07 does not engage hinge residues. Consequently, MPPT-K07 retained uncompromised potency against palbociclib-resistant cell lines harboring gatekeeper mutations (CDK4^{F93L}, IC₅₀ = 4.6 nM).



5. CONCLUSIONS & TRANSLATIONAL PRAXIS

In conclusion, our study demonstrates that integrating multi-task generative transformers with active-learning molecular dynamics achieves an unprecedented acceleration of precision medicinal chemistry. Lead compound MPPT-K07 combines picomolar allosteric affinity, remarkable kinome selectivity, and favorable oral bioavailability (F = 68%), establishing an auspicious candidate for clinical development in refractory metastatic malignancies.

Ethical Compliance & Standards: All research and peer-review protocols were conducted in strict concordance with the Committee on Publication Ethics (COPE) core guidelines and UGC-CARE editorial benchmarks. In vivo animal experiments were approved by the Institutional Animal Ethics Committee (IAEC Protocol #MPPT-2026-04).

Data & Code Availability: Model weights, tokenizers, cryo-EM pocket coordinates, and molecular dynamics trajectories are deposited under the MIT License on Zenodo (DOI: 10.5281/zenodo.10849201) and mirrored at the MPPT Journal Open Science repository (mpptjournal.com).

Author Contributions (CRediT): Conceptualization: P.P., A.S.; Methodology & Machine Learning: A.S., E.R.; Chemical Synthesis: V.N.; Investigation & In Vitro Assays: A.S., E.R.; Supervision & Project Administration: P.P., V.N.; Writing – original draft: A.S., P.P.; Writing – review & editing: All authors.

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Conflicts of Interest: The authors declare no competing financial or personal interests.

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